

< Previous



Next >

Prototype-based models

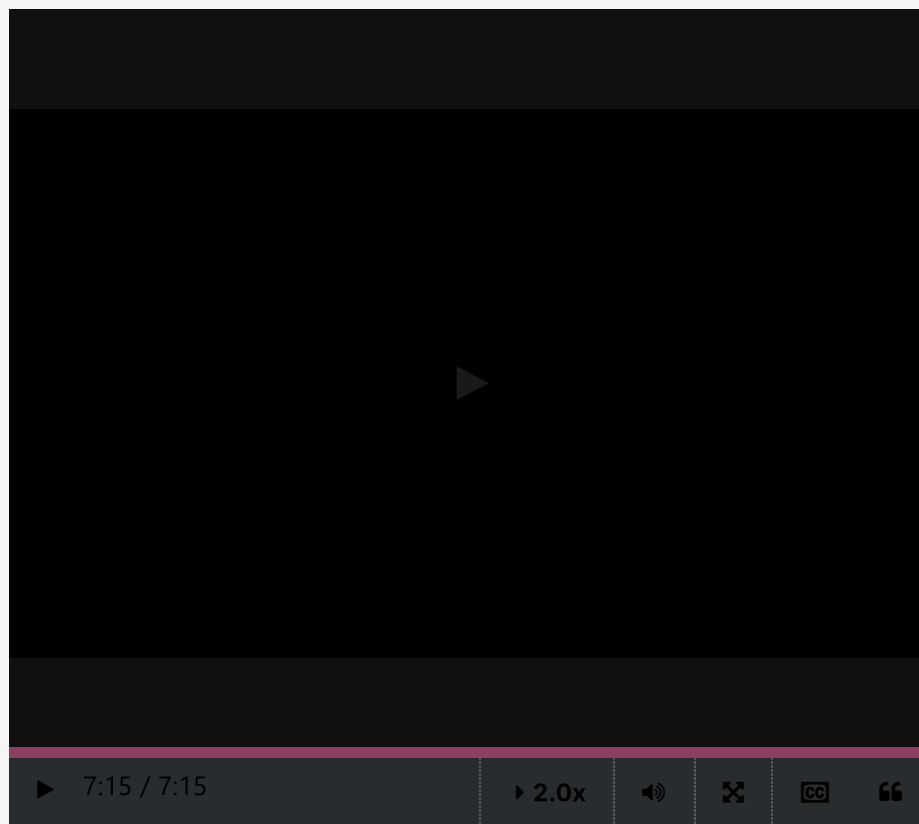
 Bookmark this page

Question: Does learning really amount to a compression task?

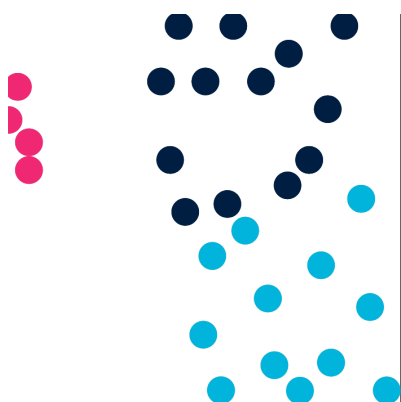
Answer: Yes, definitely!

Prototype-based models are one of the most popular machine learning techniques. They offer a nice illustration of the idea that learning amounts to achieving compression.

Clustering: From compression to MML

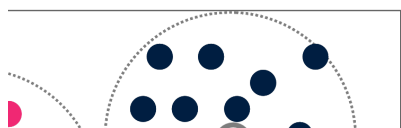


we can run a K-means algorithm with increasing numbers of clusters: the first run will yield two clusters, the second run three clusters, and so on. If we compute the complexity, we see that the minimum value is reached for the number of clusters which seems the most appropriate. Conclusion: Clustering can be understood



1

fication task as an **ed learning**. Associate a label (*e.g.* h data point.



Prototype-based models 1

1/1 point (ungraded)

In what way do *unsupervised* prototype-based models achieve compression?

- ☒ Because distances to prototypes are small.
- ☐ Because one gets rid of data points and only keeps prototypes.

☐ Because prototypes are kept far from each other.



Answer

Correct:

The distances of data points to their closest prototype are smaller than absolute distances.

Submit

You have used 1 of 1 attempt

i Answers are displayed within the problem

Prototype-based models 2

1/1 point (ungraded)

In what way does the *supervised* learning of classification achieve compression?

☐ Because distances to prototypes are small.

☒ Because one gets rid of data points and only keeps prototypes.

☐ Because prototypes are kept far from each other.



Answer

Correct:

Prototypes suffice to decide on labels, so we can forget about data points and only keep prototypes (as compared to rote learning in which the decision function must keep all data points).

Submit

You have used 1 of 1 attempt

i Answers are displayed within the problem

< Previous

Next >

Some Rights Reserved



edX

[About](#)

[Affiliates](#)

[edX for Business](#)

[Open edX](#)

[Careers](#)

[News](#)

Legal

[Terms of Service & Honor Code](#)

[Privacy Policy](#)
[Accessibility Policy](#)
[Trademark Policy](#)
[Sitemap](#)
[Cookie Policy](#)
[Your Privacy Choices](#)

Connect

[Idea Hub](#)
[Contact Us](#)
[Help Center](#)
[Security](#)
[Media Kit](#)

